Hybrid Balance Control for Wheeled Bipedal Robot via Residual Reinforcement Learning Optimization

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***Abstract*—Balance control for wheeled bipedal robots (WBRs) is a challenging task due to their nonlinear dynamics and underactuated characteristics. Traditional feedback controllers, while capable of achieving stability, become overly complex and difficult to design as the robot's degrees of freedom increase. Moreover, inaccurate system models directly affect control performance. To address these challenges, reinforcement learning (RL) offers a data-driven approach. However, when applied directly to WBRs, the training process is time-consuming and suffers from poor convergence due to their inherent static instability. Therefore, a hybrid balance control method optimized by residual RL is proposed to improve the balance performance of the robots. Firstly, a basic balance controller and a pose controller are established based on the model. Then the hybrid RL control (HRC) framework is designed, which performs coupled optimization of all degrees of freedom by real-time online compensation of control torques, improving motion stability and coordination. By designing terminal task-oriented reward functions and diverse training tasks, the policy's generalization ability across various tracking scenarios is enhanced. In addition, to improve the robustness of the policy against complex terrains and external disturbances, random factors such as environmental uncertainty and variability of physical parameters are introduced during training. Finally, sufficient physical simulation results are provided to demonstrate the superior performance of the proposed method in terms of tracking accuracy, generalization, and robustness.**

***Index Terms*—Wheeled bipedal robot, balance control, residual reinforcement learning.**

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I. Introduction

Wheeled-legged robots (WBRs) combine wheel speed with legged adaptability, enabling strong mobility in complex environments. They are compact, flexible, and low-cost, making them ideal for tasks like rescue [1], surveying [2], and logistics [3], and attracting much research interest [4]- [8]. However, their underactuated, nonlinear nature makes balance control a major challenge essential for motion and terrain adaptability.

Traditional model-based methods typically use a simplified second-order inverted pendulum model and design controllers such as Linear Quadratic Regulator (LQR) [4] or Model Predictive Control (MPC) [5] to achieve balance. Furthermore, leg pose control is enhanced by introducing decoupling control [6], dynamic compensation [7], and non-holonomic constraints [8]. Although the above traditional control strategies have been widely applied, their performance is severely constrained by the accuracy of system models. Further, when the robot has a high number of joint degrees of freedom, the controller design becomes particularly complex.

To overcome the limitations of traditional controllers, deep reinforcement learning (DRL) provides a data-driven approach. It learns an end-to-end mapping from perception to action through interacting with the environment [9], [10], without requiring an accurate system model. This makes it better suited for handling complex and dynamic robotic tasks. However, directly using standard DRL algorithms for training typically requires a large amount of data [11], making it a lengthy interaction process. Furthermore, the theoretical assurance about algorithmic convergence and system stability

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in DRL is absent, which leads to high safety risks during training [12], [13]. Specifically, the WBR has inherent static instability, which may lead it to deviate from the desired state during the initial exploration phase, making it difficult to find an effective control strategy. This high exploration cost and low reward training process significantly reduces the algorithm's convergence speed, which even leads to training failure.

Introducing prior knowledge into DRL training is effective to improve sample efficiency and stabilize the training [14], [15]. Residual RL [16] offers a flexible approach by decomposing the policy into two modules: a baseline policy and a residual policy. The baseline policy, which introduces prior knowledge, provides an initial, relatively good strategy. Subsequently, the residual policy concentrates on learning how to improve this baseline policy to achieve better performance [17]. Specifically, Baek et al. [18] successfully apply it to the WBR and achieve great improvement in balancing and tracking. Unfortunately, the residual policy is constrained to compensating only for wheel actions on flat ground, without considering leg actions or uneven terrains, which limits the training task to a relatively single scenario.

Considering the aforementioned challenges, the HRC method is proposed for the WBR in this work, which combines the advantages of traditional model-based controllers and RL. On one hand, the introduction of traditional controllers stabilizes the training process and improves sample efficiency. On the other hand, RL allows for online optimization for the traditional controllers, thereby enhancing control precision, system robustness, and generalization ability. Firstly, the mechanical structure and dynamic model of the robot are analyzed, based on which a balance controller and a pose controller are designed as the basic model-based controller (BMC). Subsequently, a novel RL framework HRC is designed, which trains a residual policy for the terminal task. It generates compensation torques for all joint motors, including all hip forward motors, knee motors and wheel motors to optimize BMC online. In contrast to previous studies that only compensated for wheel actions [18], [19], the proposed method HRC allows for more precise control of the robot's overall movement, improving balance and coordination performance. Furthermore, during the training process, several factors such as task diversity, domain randomization, and terrain variations are fully considered to enhance the robustness and generalization of the policy. Additionally, HRC relies only on the control torque information provided by BMC, without focusing on the specific design details, making it adaptable to various basic controllers and ensuring good generalization. The training framework of HRC can be seen in Fig. 1. The main contributions of this work are summarized as follows:

1) For the statically unstable WBR, a hybrid balance control framework HRC is proposed, which combines RL and traditional feedback control, addressing the challenges of complex traditional controller design and the difficulty of RL training convergence. By real-time online compensation of control torques, HRC performs coupled optimization for the robot's all six motor torques, generating more stable movements.
2) By designing terminal task-oriented reward functions, HRC effectively optimizes different basic control torques, demonstrating good coordination performance and adaptability to different basic controllers. The diverse training task setup further improves the policy's generalization ability, making it applicable to various tracking scenarios.
3) To enhance the robustness of the policy, environment uncertainty and variations in physical parameters are introduced. This enables the robot to maintain satisfactory tracking accuracy when facing terrains with different slopes, as well as remain balanced and quickly recover stability when subjected to external disturbances.

II. System Model and Basic Controller

This section presents the robot's mechanical structure, dynamic model, and derives the forward and inverse kinematics equations. A basic balance controller and a pose controller are then designed as the BMC. This module initially stabilizes the system, providing a reliable initial state for subsequent RL policy training.

A. System Kinematics

Fig. 2 shows the mechanical structure and kinematic model of the robot. The local control frame $\{C\}$ has its origin at the midpoint of the line connecting the two wheels, with the x-axis pointing forward horizontally and coinciding with the positive direction of the robot's movement. The body frame $\{B\}$ is fixed on the floating base of the WBR, coinciding with the center of the IMU, which serves as the origin for connecting all joint frames. COG represents the center of gravity of the robot relative to $\{C\}$, denoted as $\boldsymbol{p}_c = [x_c,\ y_c,\ z_c]^\top$. Based on the coordinate definition shown in Fig. 2, the forward kinematics are derived, that are, the homogeneous transformation matrices $\boldsymbol{T}$ from $\{C\}$ to every joint coordinate. With this transformation, precise control of the robot's joints can be achieved by applying control commands in the control frame $\{C\}$.

Define $\boldsymbol{q} = [\theta_{1l}, \theta_{1r}, \theta_{2l}, \theta_{2r}, \theta_{3l}, \theta_{3r}, \alpha, \beta, \gamma]^\top$, where α, β, γ represent the roll, pitch and yaw angle of the robot, respectively. Based on the transformation relationship between the frames, the expression for $\boldsymbol{p}_c$ can be derived:

$$\boldsymbol{p}_c(q) = \frac{\sum m_i\ {}^C\boldsymbol{p}_i(q)}{\sum m_i} \tag{1}$$

where ${}^C\boldsymbol{p}_i$ represents the i-th link's center of mass position in $\{C\}$, m_i represents the mass of the i-th link. Further, the Jacobian matrix for inverse kinematics of the joint position can be derived:

$$\boldsymbol{J} = \begin{bmatrix} \frac{\partial x_c}{\partial \theta_{1l}} & \frac{\partial x_c}{\partial \theta_{1r}} & \cdots & \frac{\partial x_c}{\partial \beta} & \frac{\partial x_c}{\partial \gamma} \\ \frac{\partial y_c}{\partial \theta_{1l}} & \frac{\partial y_c}{\partial \theta_{1r}} & \cdots & \frac{\partial y_c}{\partial \beta} & \frac{\partial y_c}{\partial \gamma} \\ \frac{\partial z_c}{\partial \theta_{1l}} & \frac{\partial z_c}{\partial \theta_{1r}} & \cdots & \frac{\partial z_c}{\partial \beta} & \frac{\partial z_c}{\partial \gamma} \end{bmatrix} \in \mathbb{R}^{3\times 9} \tag{2}$$

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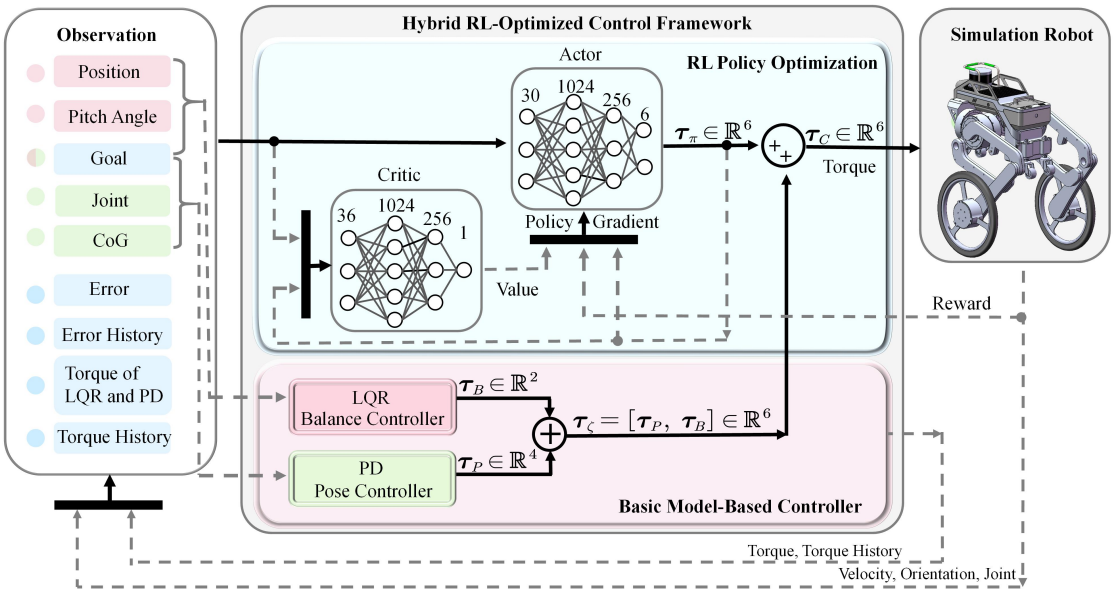


Fig. 1. The hybrid RL control training framework of the WBR. It consists of two modules: the basic model-based controller that ensures system stability, and the RL policy that optimizes control performance in real-time. Both directly generate control torques, which are subsequently summed and applied to the robot joint motors, as shown by the solid lines in the diagram.

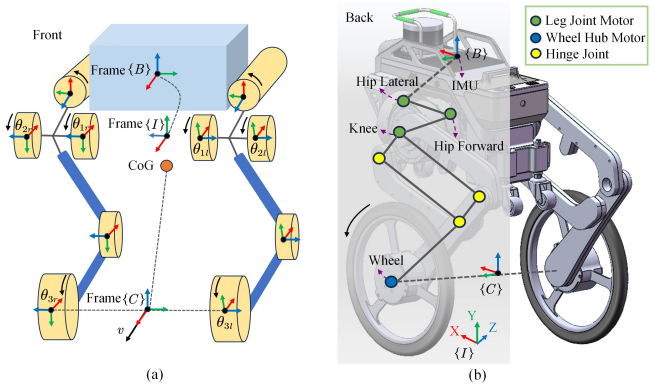


Fig. 2. (a) Kinematic model and coordinate system definition of the WBR. (b) Mechanical structure and motor distribution of the WBR. The robot has eight motors in total, with green circles representing the leg motors and blue circles indicating the wheel hub motors. The hip lateral motors allow for the abduction and adduction of the legs. The hip forward and knee motors coordinate to control the pose of the floating base. In this work, the rotation of the hip lateral motors is not considered. θ_{1l}, θ_{1r}, θ_{2l}, θ_{2r}, θ_{3l} and θ_{3r} represent the rotational angles of the hip forward motors, knee motors, and wheel hub motors, respectively, which are directly measured by encoders. The euler parameters of the floating base are measured by the built-in inertial measurement unit (IMU).

B. Basic Model-Based Controller

The BMC of the WBR consists of two modules: a balance controller and a pose controller. The balance controller controls the wheel hub motors to maintain the robot's balance, with the torques denoted as $\boldsymbol{\tau}_B \in \mathbb{R}^2$. The pose controller adjusts the floating base's pose by controlling the hip forward motors and knee motors, with the torques denoted as $\boldsymbol{\tau}_P \in \mathbb{R}^4$. To highlight the optimization capability of the proposed HRC framework and avoid introducing excessive computational complexity, the design of the BMC in this section is intentionally simplified, requiring only an initial stable control strategy.

1) Basic Balance Controller: Simplify the robot into a classical inverted pendulum model. The second-order dynamic model of the system is established based on the Euler-Lagrange equation:

$$\begin{aligned} a\ddot{x}_w + b\ddot{\theta}_p - m_p R l sin\left(\theta_p\right)\dot{\theta}_p^2 &= u \\ b\ddot{x}_w + m_p l^2\ddot{\theta}_p - m_p g l sin\theta_p &= -u \end{aligned} \tag{3}$$

where $a = mR^2 + m_pR^2 + j$, $b = m_pRlcos\left(\theta_p\right)$. l is the height of the robot. m_p is the mass of the pendulum, m is the mass of the wheel. θ_p is the inclination angle between the vertical direction and the virtual pole. u is the control input, and j is rotational inertia of the hub motor.

The LQR is chosen as the basic balance controller, which is an optimized linear feedback controller. By minimizing a quadratic performance index, it allows the system to achieve the desired state while balancing the system state and control input. The optimal feedback gain matrix $\boldsymbol{K}$ is determined by tuning the weight matrices $\boldsymbol{Q}$ and $\boldsymbol{R}$, and solving the Riccati equation. Multiplying the error between the current state $\boldsymbol{q}_b = \left[x_w, \dot{x}_w, \theta_p, \dot{\theta}_p\right]^\top$ and the desired state $\boldsymbol{q}_{bd}$ by the gain matrix $\boldsymbol{K}$, the optimal control input $\boldsymbol{u}$ is obtained:

$$\boldsymbol{u} = -\boldsymbol{K}\left(\boldsymbol{q}_b - \boldsymbol{q}_{bd}\right) \tag{4}$$

Linearizing (3) around the equilibrium point and adjust the weight matrices Q and R, the optimal gain is obtained as $\boldsymbol{K}^* = [14, 10, 90, 10]$. The adjustment of yaw angle γ is integrated into the LQR design.

2) Basic Pose Controller: The dynamic pose control of the robot's floating-base is achieved by regulating the hip motors and knee motors using Proportional-Derivative (PD) controllers. The control objectives are to track the CoG height, adjust the roll and pitch angles, and prevent lateral wheel

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slip. The last two terms are represented by $\boldsymbol{\psi}$ and $\boldsymbol{\chi}$, respectively. Define the vectors $\boldsymbol{q}_\theta = [\theta_{1l}, \theta_{1r}, \theta_{2l}, \theta_{2r}, \theta_{3l}, \theta_{3r}]^\top$ and $\boldsymbol{q}_\psi = [\alpha, \beta, \gamma]^\top$. The control objectives can be organized as:

$$\begin{aligned} \dot{z}_c &= \boldsymbol{J}_{z\theta}\dot{\boldsymbol{q}}_\theta + \boldsymbol{J}_{z\psi}\dot{\boldsymbol{q}}_\psi \\ \dot{\boldsymbol{\psi}} &= \boldsymbol{J}_{\psi\theta}\dot{\boldsymbol{q}}_\theta + \boldsymbol{J}_{\psi\psi}\dot{\boldsymbol{q}}_\psi \\ \dot{\boldsymbol{\chi}} &= \boldsymbol{J}_{\chi\theta}\dot{\boldsymbol{q}}_\theta + \boldsymbol{J}_{\chi\psi}\dot{\boldsymbol{q}}_\psi \end{aligned} \tag{5}$$

Through inverse kinematics, the desired velocity of joints $\dot{\boldsymbol{q}}_{\theta d}$ can be derived from the target velocity of the terminal task:

$$\dot{\boldsymbol{q}}_{\theta d} = \begin{bmatrix} \boldsymbol{J}_{z\theta} \\ \boldsymbol{J}_{\psi\theta} \\ \boldsymbol{J}_{\chi\theta} \end{bmatrix}^+ \left(\begin{bmatrix} \dot{z}_{cd} \\ \dot{\boldsymbol{\psi}}_d \\ \dot{\boldsymbol{\chi}}_d \end{bmatrix} - \begin{bmatrix} \boldsymbol{J}_{z\psi}\dot{\boldsymbol{q}}_\psi \\ \boldsymbol{J}_{\psi\psi}\dot{\boldsymbol{q}}_\psi \\ \boldsymbol{J}_{\chi\psi}\dot{\boldsymbol{q}}_\psi \end{bmatrix} \right) \tag{6}$$

The subscript θ corresponds to $\boldsymbol{q}_\theta$, subscript ψ corresponds to $\boldsymbol{q}_\psi$, and subscript d represents the desired value. By integrating $\dot{\boldsymbol{q}}_{\theta d}$, the desired joint position $\boldsymbol{q}_{\theta d}$ can be obtained. The PD parameters for the hip forward motors are set to $k_{p1} = 85$, $k_{d1} = 12$, and $k_{p2} = 75$, $k_{d2} = 15$ for the knee motors. Combining the balance controller and the pose controller, the control torques of the BMC are obtained as $\boldsymbol{\tau}_\zeta = [\boldsymbol{\tau}_P, \boldsymbol{\tau}_B] \in \mathbb{R}^6$.

III. Hybrid Reinforcement Learning Optimization

While the BMC ensures stable motion, its performance may degrade under parameter variations or unexpected terrain. Additionally, insufficient coordination with the pose controller can lead to inefficient or unnatural movements. In this section, a residual RL strategy is trained to optimize the controll performance.

A. State Space

State $\boldsymbol{s}$ at time t is defined as $\boldsymbol{s}_t = [\boldsymbol{s}^r, \boldsymbol{s}^g, \boldsymbol{s}^e, \boldsymbol{s}^\tau]$. The detailed definitions of each component are given below.

- Robot state vector $\boldsymbol{s}^r = \left[\boldsymbol{q}_b^t, \boldsymbol{p}_c^t, \dot{\boldsymbol{p}}_c^t, \boldsymbol{q}_\theta^t, \dot{\boldsymbol{q}}_\theta^t\right]$.
- Desired balance state and desired pose $\boldsymbol{s}^g = \left[\boldsymbol{q}_{bd}^t, z_{cd}^t, \dot{z}_{cd}^t, \boldsymbol{q}_{\psi d}^t\right]$. Specifically, to enhance the policy's robustness against unknown terrains, the inaccurate desired inclination angle θ_{pd} information is intentionally removed. This aims to encourage the agent to learn from other state information, enabling it to make accurate decisions when facing terrains with unknown angles.
- Motion error and history $\boldsymbol{s}^e = \left[\boldsymbol{e}^t, \boldsymbol{e}^{t-1}\right]$, which is calculated based on $\boldsymbol{s}^g$.
- Torque and history $\boldsymbol{s}^\tau = \left[\boldsymbol{\tau}_\zeta^t, \boldsymbol{\tau}_\zeta^{t-1}, \boldsymbol{\tau}_\pi^{t-1}, \boldsymbol{\tau}_C^{t-1}\right]$, where $\boldsymbol{\tau}_\pi$ represents the torques generated by the RL optimization strategy and $\boldsymbol{\tau}_C$ represents the final control torques.

B. Action Space

The RL optimization policy directly generates torques $\boldsymbol{\tau}_\pi \in \mathbb{R}^6$, which compensate to all six motors based on the BMC. The final torques applied to each motor are $\boldsymbol{\tau}_C = \boldsymbol{\tau}_\zeta + \boldsymbol{\tau}_\pi$. The Soft Actor-Critic (SAC) algorithm, a RL method based on stochastic policy, is used to train the policy network. By introducing entropy, SAC encourages the agent to explore a wider range of state-action pairs, preventing converging to local optima. The stochastic exploration strategy effectively balances exploration and exploitation, improving learning efficiency and ultimately obtaining a better policy.

It is worth noting that the proposed HRC framework interacts with the BMC only through torque information, and the design of the state and action space is independent of the specific structure of the BMC. This modular design allows the framework to be flexibly integrated into other model-based control methods, demonstrating good generalization.

C. Reward Function

To achieve smooth tracking of the target and maintain the desired pose of floating base, the reward function is defined as: $r = r_{\theta lim} + r_{bp}$. By penalizing actions that exceed the joint limits of the hip and knee motors, $r_{\theta lim}$ guides the robot to learn a safe range of motion. r_{bp} represents the balance and pose reward, optimizing the terminal task performance. In this reward function design, only maximum joint limit is considered for each joint's state. The joint states are not utilized to guide task optimization.

r_{bp} guides optimization by measuring the deviation between the real terminal state and the desired state $\boldsymbol{s}^g$, expressed as $r_{bp} = r_b + r_z + r_\psi$. r_b measures the balance state $\boldsymbol{q}_b$, encouraging the robot to minimize tracking errors and maintain floating base stable. r_z encourages the robot to maintain the desired CoG height z_c. r_ψ measures the robot's pose performance $\boldsymbol{q}_\psi$, ensuring the robot to maintain the desired motion posture. All error terms measured in r_{bp} correspond sequentially to the elements in $\boldsymbol{e}^t$ and are quantified using the function $f(x) = 1/(\exp(x) + \exp(-x) + 2) - 2$. Specifically, to enable the policy to better adapt to unknown terrains, this work does not set a specific desired inclination angle value θ_{pd}, but enhances system stability by limiting its rate of change $\dot{\theta}_p$.

Instead of guiding individual joint movements, this work adopts a more global reward function design that directly evaluates the robot's performance based on the completion of the terminal task. By rewarding balance and pose performance, the robot is encouraged to make decisions that optimize the overall task, rather than focusing on local joint movements. This design provides the robot with greater exploration freedom and enables it to learn more effective action strategies that comprehensively consider global objectives. Additionally, the terminal task-oriented reward design can avoid local optima that may arise in traditional methods, enhancing the robot's adaptability to diverse and dynamic environment.

D. Domain Randomization

To enhance the model's robustness against environmental uncertainties and variations in model parameters, a domain randomization strategy is employed during training. Firstly, random noise is introduced to physical parameters like robot module mass to simulate modeling errors. Additionally, add random noise to joint position sensor and IMU to simulate more realistic sensor equipment. Furthermore, environmental parameters such as ground friction and slope angle are randomly generated to improve the model's adaptability to diverse terrain.

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Algorithm 1 Hybrid RL Control (HRC) Algorithm

Input:
policy network $\boldsymbol{\pi}$, replay buffer B,
basic model-based controller (BMC)

for $epoch$ = 1, $\cdots$, $n_epoches$ **do**
  for $episode$ = 1, $\cdots$, $n_episodes$ **do**
    get initial state $\boldsymbol{s}^1$
    for t = 1, $\cdots$, max_steps **do**
      get control torques $\boldsymbol{\tau}_\zeta$ based on BMC
      get action $\boldsymbol{\tau}_\pi$ based on policy $\boldsymbol{\pi}$
      execute action $\boldsymbol{\tau}_C = \boldsymbol{\tau}_\zeta + \boldsymbol{\tau}_\pi$
      obtain reward r^t, new state $\boldsymbol{s}^{t+1}$
      store $\left\{\boldsymbol{s}^t, \boldsymbol{a}^t, r^t, \boldsymbol{s}^{t+1}\right\}$ in B
    end for
    update the policy $\boldsymbol{\pi}$ by SAC
  end for
  evaluate with the current policy $\boldsymbol{\pi}$
end for

E. Network Structure

The policy network employs a two-layer fully connected architecture with 1024 and 512 neurons, respectively, using the tanh activation function. The network takes a 30-dimensional state vector as input and outputs a 6-dimensional action vector. Training is conducted using the SAC algorithm, with a iteration of maximum 5000 steps. The policy network is updated every 1000 steps, and the batch size is set to 512. The learning rate is 1e-3, the discount factor is 0.99, the entropy regularization coefficient is initially set to 0.1, and the reward scale is 10. The algorithm is trained on a computer equipped with a 12-core CPU and a NVIDIA 3060 GPU. The detailed procedure is described in Algorithm 1.

IV. Physical Simulation Results and Discussion

In this section, the proposed HRC framework is validated in the Webots simulation environment using a robot model consistent with the real device in structure and physical parameters.

A. Training and Performance

To evaluate the effect of the proposed method, two groups of training scenarios are designed:

- Group 1: Keep the height constant, set random movement target ranging from 1 m to 4 m, and consider slope with random angles of 0 deg, 5 deg, and 10 deg. Additionly, maintain the pose angles at $\boldsymbol{q}_{\psi d} = \mathbf{0}$.
- Group 2: Set random movement target ranging from 1 m to 4 m on flat ground, with random CoG height change target ranging from 0.05 m to 0.15 m. Additionly, maintain the pose angles at $\boldsymbol{q}_{\psi d} = \mathbf{0}$.

The first group primarily focuses on the robot's balance ability when facing unknown terrains. The second group further examines the robot's comprehensive balance performance when simultaneously dealing with horizontal movement and CoG height change. The proposed HRC method is compared with the following controllers:

- Basic model-based controller (BMC) only.
- Model-free RL algorithms only, using SAC and Deep Deterministic Policy Gradient (DDPG).
- Replace SAC in the HRC framework with DDPG, while keeping all other settings unchanged, denoted as BMC-DDPG.

Fig. 3 shows the reward variation curves for different methods under the two group scenarios. Three random seeds are set. A total of 40 epochs are trained, accumulating 2 million sampled data. Results show that the proposed HRC method converges at around 20 epochs, demonstrating the best performance. In contrast, methods that directly apply RL algorithms (SAC, DDPG) fail to complete the balance task with the same amount of sample data. This indicates that directly applying RL to the WBR suffers from low sample efficiency and difficulty in exploring stable strategy. Both RL-optimized methods (HRC and BMC-DDPG) are able to converge to stable strategy and outperform the BMC, indicating that the prior knowledge provided by the BMC helps stabilize RL training and accelerate convergence. At the same time, the introduction of RL helps to compensate for the adaptability shortcomings of BMC in complex environment, enhancing the overall performance of the BMC controller.

Compare the performance of HRC and BMC-DDPG. In Group 1, BMC-DDPG demonstrates premature convergence around the 10th epoch, getting stuck in a local optimum. The HRC method exhibits larger fluctuations in the early stage but ultimately converges to a superior policy. In Group 2, as the task difficulty increases, the learning process for BMC-DDPG becomes more challenging, resulting in slight strategy improvement. In contrast, HRC exhibits robust convergence and achieves significantly higher rewards. This suggests that the random strategy search of SAC is more suitable for handling complex end-to-end control tasks compared to the deterministic strategy search of DDPG. The random strategy, by generating action probability distributions and performing random sampling, helps the agent explore a broader optimization space, thereby facilitating the discovery of globally optimal policies.

Two tasks are set to demonstrate the specific performance of the proposed strategy:

- Task 1: Move to a 2.5 m target, while maintaining a constant CoG height and stable pose.
- Task 2: Move to a 2.5 m target, while raising the CoG height by 0.12 m and maintaining stable pose.

The performance of Task 1 on flat ground is shown in Fig. 4, and the task demonstration is shown in Fig. 6a. Since SAC and DDPG fail to train stable motion policies, their results are not presented. During the tracking process, the roll angle and yaw angle of the body are both maintained within 0.001 rad. To measure the error between the target values and the actual values of the position x_w, velocity v, inclination angle θ_p,

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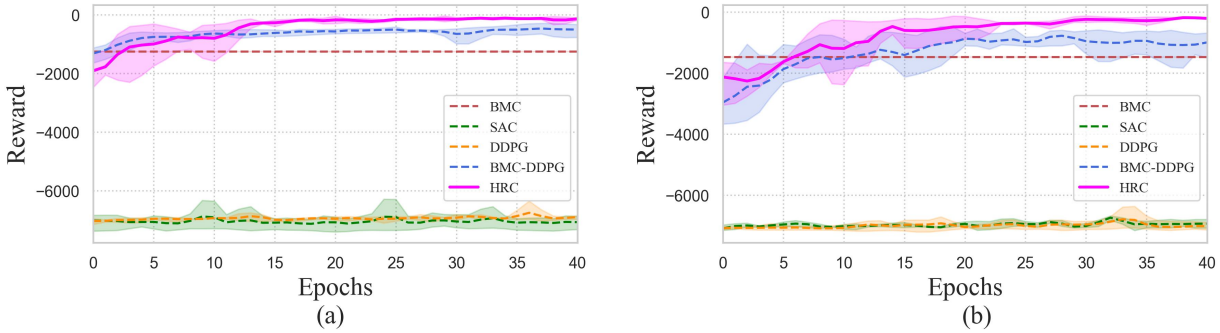


Fig. 3. Comparison of the mean reward between the BMC, SAC, DDPG, BMC-DDPG and HRC during training process. (a) Group 1. (b) Group 2.

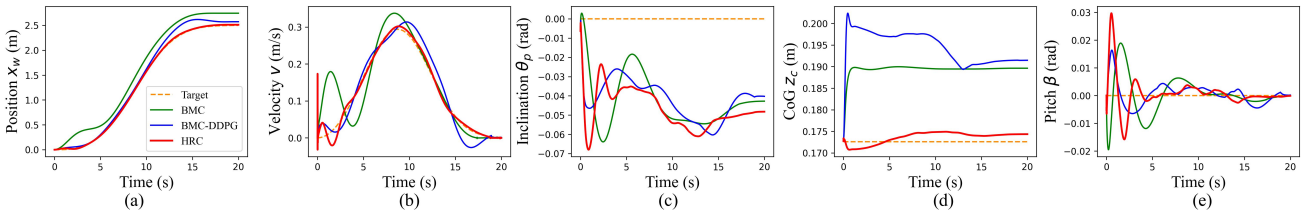


Fig. 4. Performance of Task 1 on flat ground: (a) Position x_w change curve. (b) Velocity v change curve. (c) Inclination angle θ_p change curve. (d) CoG height z_c change curve. (e) Pitch angle β change curve.

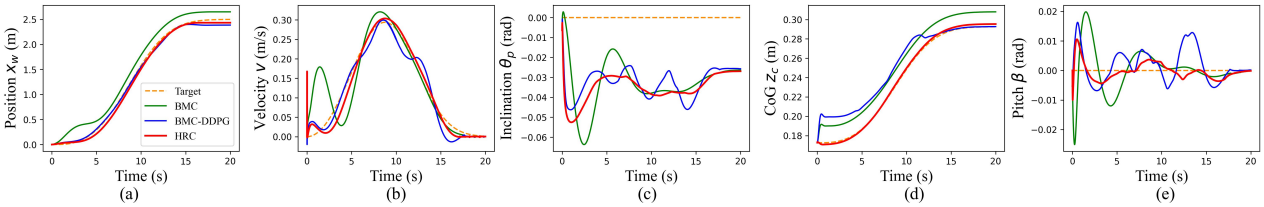


Fig. 5. Performance of Task 2 on flat ground: (a) Position x_w change curve. (b) Velocity v change curve. (c) Inclination angle θ_p change curve. (d) CoG height z_c change curve. (e) Pitch angle β change curve.

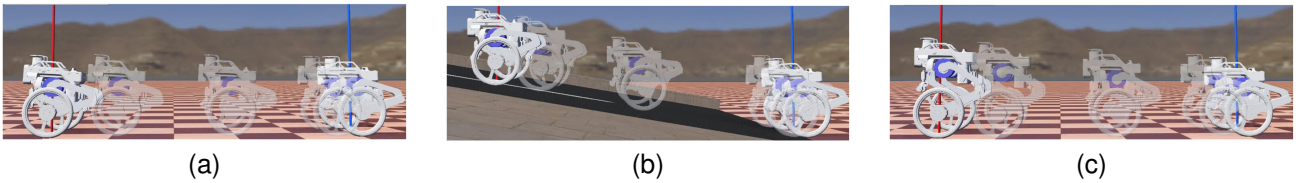


Fig. 6. Task demonstration. (a) Task 1 on flat ground. (b) Task 1 on slope with 10 deg. (c) Task 2 on flat ground.

TABLE I
RMSE RESULTS WITH DIFFERENT TASKS.

Task	Slope	Method	Metric(RMSE)				
			x_w(m)	v(m/s)	θ_p(rad)	z_c(m)	β(rad)
Task 1	0 deg	BMC	0.2475	0.0533	0.0157	0.0168	0.0065
		BMC-DDPG	0.0966	0.0265	0.0128	0.0222	**0.0039**
		HRC	**0.0134**	**0.0119**	**0.0106**	**0.0032**	0.0057
Task 1	5 deg	BMC	0.1397	0.0400	0.0177	0.0167	0.0064
		BMC-DDPG	0.1032	0.0127	0.0134	0.0051	0.0049
		HRC	**0.0401**	**0.0099**	**0.0114**	**0.0015**	**0.0042**
Task 1	10 deg	BMC	0.3436	0.0619	0.0461	0.0170	0.0106
		BMC-DDPG	0.1193	0.0125	0.0313	0.0023	0.0128
		HRC	**0.0366**	**0.0086**	**0.0289**	**0.0012**	**0.0103**
Task 2	0 deg	BMC	0.1909	0.0522	0.0123	0.0161	0.0069
		BMC-DDPG	**0.0548**	0.0225	0.0084	0.0175	0.0058
		HRC	0.0578	**0.0176**	**0.0075**	**0.0018**	**0.0028**

CoG height z_c, and base pitch angle β, the root mean square error (RMSE) is used as the evaluation metric, shown in Table I. Compared to BMC, the proposed HRC framework achieves a 94% improvement in average performance for the position x_w tracking, reducing the steady-state error from 0.235 m to 0.013 m. The position tracking curve of HRC almost coincides with the target curve. The average performance on height z_c maintaining improves by 80%, and the steady-state error reduces from 0.017 m to 0.0028 m. The stability of θ_p and β has also been slightly optimized. In contrast, BMC-DDPG does not find an overall optimal strategy. Although there is great optimization in x_w tracking, with the steady-state error reduced to 0.06 m, it performs poorly in maintaining height z_c stable.

The performance of Task 2 on flat ground is shown in Fig. 5 and Table I, with the task demonstration in Fig. 6c. Compared to Task 1, Task 2 has a higher demand on the controller's coordination ability. HRC not only effectively compensates for

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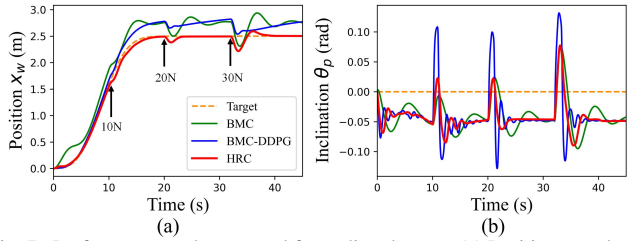


Fig. 7. Performance under external force disturbances. (a) Position x_w change curve. (b) Inclination angle θ_p change curve.

position x_w and height z_c error, but also significantly improves the pose stability. Compared to BMC, the variations in θ_p and β of HRC are reduced by 30% and 59%, respectively, and the corresponding curves in Fig. 5 are greatly smoother. In contrast, although BMC-DDPG reduces the tracking errors, the motion is more fluctuant and the stability is poorer.

B. Test 1: Slope and External Force Robustness

In this section, the robustness of the proposed strategy is tested under slope terrains and external disturbances. During the tests, all control strategies are kept blind to the specific information about the slope angles and disturbance values. Firstly, the slope terrain test is conducted. Test the Task 1 on slope of 5 deg and 10 deg, with the results shown in Table I and the task demonstration shown in Fig. 6b. As the slope angle increases, it becomes increasingly difficult to simultaneously achieve accurate position tracking and maintain base stable. However, the proposed HRC demonstrates outstanding robustness. On the slope with 10 deg, the RMSE error of x_w with BMC even reaches 0.3436 m, while HRC reduces it by 89%. The final steady-state x_w error decreased from 0.39 m to 0.03 m, and the θ_p variation is also reduced.

The external disturbance test is set as: the robot executes Task 1 on flat ground, and instantaneous external force of 10 N, 20 N, and 30 N are applied at 12 s, 20 s, and 32 s, respectively. As shown in Fig. 7, whether during movement or staying still, the HRC can quickly recover to the target trajectory and effectively suppress inclination angle oscillation under reverse external disturbances. In contrast, the BMC method exhibites significant oscillations in both θ_p and x_w, struggling to recover stability within a short period. When the force of 20 N applied, HRC experiences a 0.2 m position offset and returns to the target position within 4 s. While ensuring position accuracy, the adjustment amplitude of θ_p is relatively large, but it can recover to the stable state without fluctuations within 4 s. In comparison, the x_w offset of BMC reaches 0.5 m. Both x_w and θ_p cannot recover stability within 12 s. When the external force increases to 30 N, HRC still manages to recover quickly, while the oscillations of the BMC becomes even more severe.

C. Test 2: Trajectory Generalization

Three sets of tests are designed in this section, with the first two focusing on evaluating trajectory tracking performance and the third on height tracking capability. Fig. 8 presents the results of the trajectory tracking tests. The two trajectories

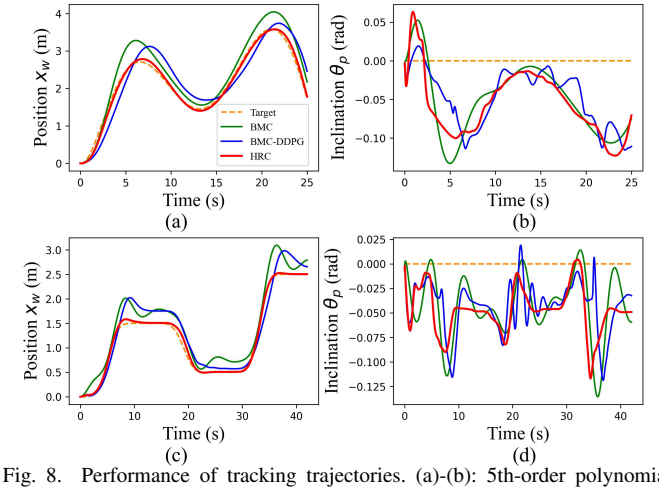


Fig. 8. Performance of tracking trajectories. (a)-(b): 5th-order polynomial random point fitting trajectory. (c)-(d): High-dynamic multi-target trajectory.

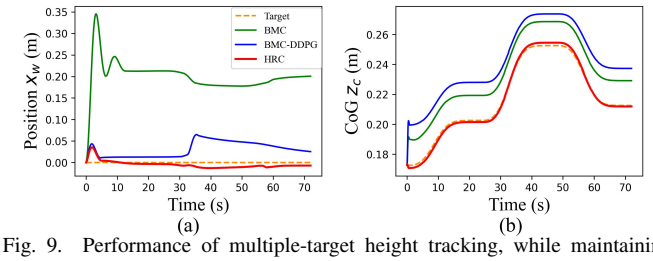


Fig. 9. Performance of multiple-target height tracking, while maintaining position stable. (a) Position x_w change curve. (b) CoG height z_c change curve.

include a 5th-order polynomial random point fitting trajectory and a high-dynamic multi-target trajectory, exhibiting stronger randomness and higher velocity variation rates compared to the training reference trajectories. As shown in the figure, the proposed HRC method can still maintain high accuracy when tracking these complex trajectories. Especially, the stability advantage of the θ_p curve in Fig. 8 is particularly significant. In contrast, the BMC and BMC-DDPG show difficulties in dealing with high-rate varying trajectories, with obvious overshoot in trajectory tracking and severe inclination angle oscillations.

Fig. 9 shows the performance in the multi-target height tracking task. The CoG height reaches 0.2 m, 0.25 m, and 0.21 m in sequence, while the robot is required to remain stationary in position. The results demonstrate that the proposed HRC exhibits excellent generalization when handling different height targets. Compared to BMC, HRC achieves higher accuracy during the height tracking process, both during the ascent phase and the steady-state phase, with the RMSE of z_c reduced from 0.016 m to 0.0014 m. Meanwhile, the position stability performance is also ensured.

D. Test 3: Different BMC Parameters Generalization

This section aims to evaluate the dependence of the proposed HRC framework on the performance of BMC. In addition to the optimal LQR parameter $\boldsymbol{K}^*$ used in Section IV-A, randomly select two other sets of LQR parameters, $\boldsymbol{K}_1 = [24, 8, 50, 9]$ and $\boldsymbol{K}_2 = [35, 15, 100, 15]$, to construct basic controllers with different performance. Compared with

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TABLE II
RMSE RESULTS WITH DIFFERENT BASIC PARAMETERS WITH TASK 1.

LQR Parameter	Slope	Metric (RMSE)	Method		
			BMC	BMC-DDPG	HRC
$\boldsymbol{K}_1$=[24,8,50,9]	0 deg	x_w(m)	0.0976	0.0557	**0.0286**
		θ_p(rad)	0.0286	0.0141	**0.0101**
	10 deg	x_w(m)	0.0820	0.1032	**0.0237**
		θ_p(rad)	0.0356	**0.0155**	0.0192
$\boldsymbol{K}_2$=[35,15,100,15]	0 deg	x_w(m)	0.1251	0.1272	**0.0290**
		θ_p(rad)	0.0164	0.0114	**0.0105**
	10 deg	x_w(m)	0.2056	0.0343	**0.0201**
		θ_p(rad)	0.0453	**0.0166**	0.0174

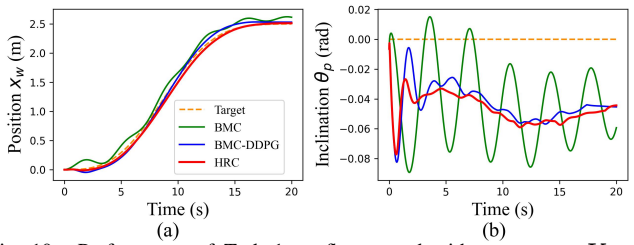


Fig. 10. Performance of Task 1 on flat ground with parameters $\boldsymbol{K}_1$. (a) Position x_w change curve. (b) Inclination angle θ_p change curve.

$\boldsymbol{K}^*$, $\boldsymbol{K}_1$ exhibits more instability and more oscillations in inclination angle. $\boldsymbol{K}_2$ shows a performance similar to $\boldsymbol{K}^*$ but with different tracking errors. Based on these two sets of parameters, the HRC policy are retrained, respectively. The same training task and hyperparameters as Group 1 in Section IV-A are maintained. The motion performance under different basic controllers are tested on flat ground and slope with 10 deg with Task 1. Select position x_w and inclination angle θ_p as indicators to evaluate the performance, with the RMSE data shown in Table II. While $\boldsymbol{K}_1$ and $\boldsymbol{K}_2$ show different performance characteristics, Table II indicates that the proposed HRC method can effectively optimize both, resulting in notable improvements in position tracking and inclination angle stability. In particular, the performance comparison of $\boldsymbol{K}_1$ on flat ground is illustrated in Fig. 10. It can be seen that, although the BMC with $\boldsymbol{K}_1$ has significant oscillations, after RL optimization, the system can recover stability and achieve high-precision tracking. Compared to BMC, the RMSE error in x_w and θ_p of HRC are reduced by 71% and 65%, respectively.

It is worth noting that in the proposed method HRC, the RL module only obtains torque information from the BMC, while its optimization objective is the pose performance in the terminal task. This indicates that the HRC framework has good generalization for different types of traditional controllers and can be applied to other model-based control schemes.

V. CONCLUSION

In this paper, a hybrid RL-optimized controller is proposed for the statically unstable WBR. By combining the advantages of RL and traditional feedback controller, it enables more robust tracking under complex terrains and varying heights. The forward and inverse kinematics models of the system are derived to design the basic model-based controller, which provides a stable initial policy. The subsequent RL optimization strategy directly generates the compensation control torques to optimize all motors, with the reward function guided by the terminal task performance. In physical simulation tests, the proposed HRC method outperforms the comparison methods, showing improved robustness when facing slope terrain and external disturbances, as well as enhanced generalization when handling irregular trajectories and multiple height target tracking. In future work, HRC will be applied to the real WBR system to further validate its performance.

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